U1 L1 Rectangular Coordinates Name: _____ HR:_____

Skills: Solve distance, midpoint, and right triangle problems.

Warm-Up: Prerequisite Skills

You'll use these three skills constantly in this unit — make sure they're solid before we start

1. Distance on a number line: $|a-b|$

Ex: distance from −3 to 5 $|5--3|=|5+3|=8$

$|-3-5|=|-8|=8$

Your turn:

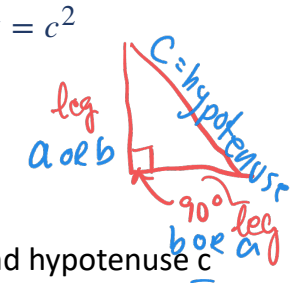
distance from −6 to 4:

$|-6-4|=|-10|=10$

$|4--6|=|4+6|=10$

2. Pythagorean Theorem

$a^2+b^2=c^2$



Your turn:

legs 6 and 8, find hypotenuse c

$a^2+b^2=c^2$

$6^2+8^2=c^2$

$36+64=c^2$

$\sqrt{100}=\sqrt{c^2}$

$c=10$

3. Simplifying a radical

Ex: $\sqrt{72}$ $3\sqrt{8}$ $3\cdot\sqrt{4\cdot2}$ $3\cdot2\sqrt{2}$ $6\sqrt{2}$

$\sqrt{9\cdot8}$ $\sqrt{9}\cdot\sqrt{8}$ $3\cdot\sqrt{8}$

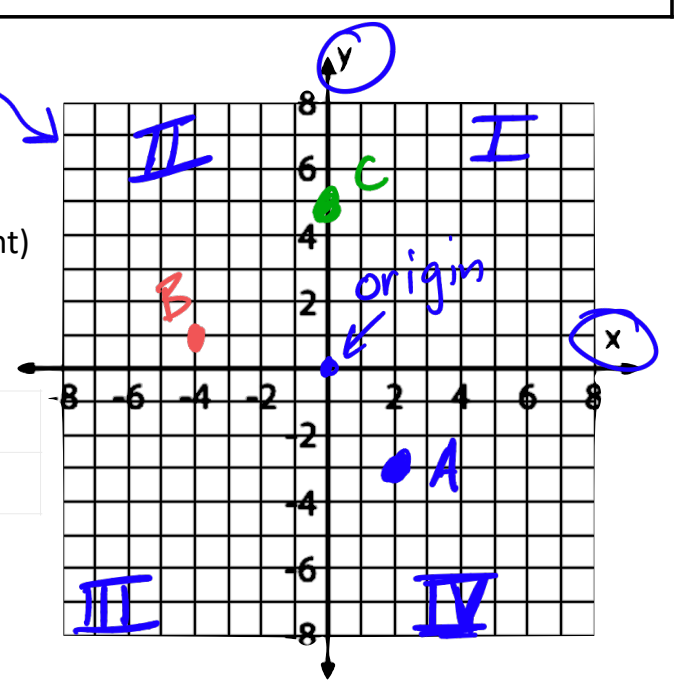
72 $36\cdot2$ $\sqrt{36\cdot2}$ $6\sqrt{2}$

$2^2=4$ $3^2=9$ $4^2=16$ $5^2=25$ $6^2=36$

Your turn: $\sqrt{20}$ $\sqrt{4\cdot5}$ $\sqrt{4}\cdot\sqrt{5}$ $2\sqrt{5}$

Vocabulary

- Cartesian plane — the coordinate grid formed by two number lines at right angles
- x-axis / y-axis — the horizontal / vertical number line
- origin — the point (0, 0) where the axes cross
- quadrants — the 4 regions the axes create (I, II, III, IV, going counter-clockwise from upper right)
- ordered pair (x, y) — x is the directed distance from the y-axis;
  y is the directed distance



The Cartesian Plane

Label each quadrant with its sign pattern, e.g. QI = (+, +).

QII: (−, +)	QI: (+ , +)
QIII: (−, −)	QIV: (+, −)

Ex: $(-1,2), (3,4), (0,0), (3,0), (-2,-3)$

You try: Graph the following and state the quadrant:

A(2, −3) quadrant: IV B(−4, 1) quadrant: II C(0, 5) quadrant: y-axis *Quick check: if x < 0 and y > 0, the point is in Quadrant* II

The Distance Formula

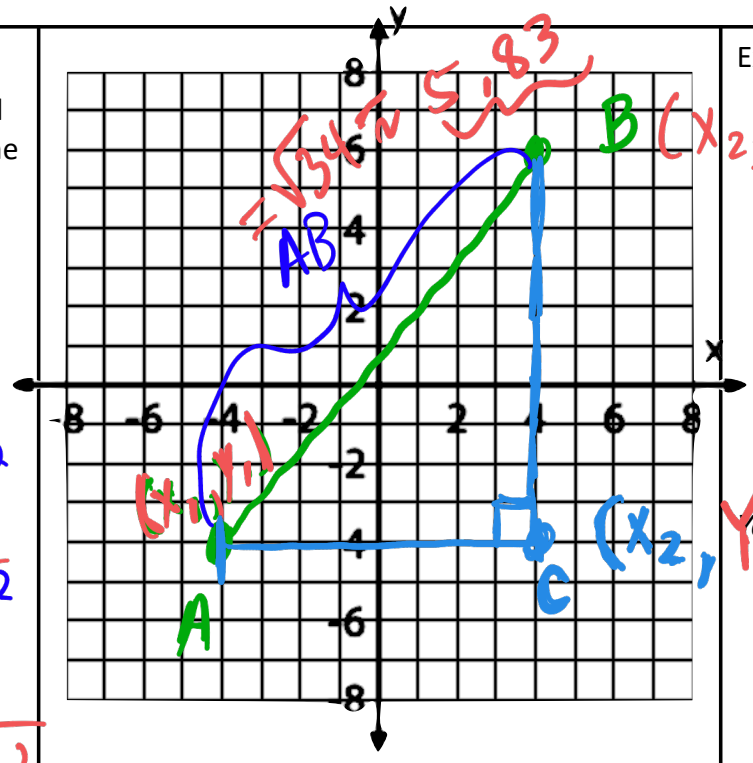
From the Pythagorean Theorem: the horizontal leg is $|x_2 - x_1|$, the vertical leg is $|y_2 - y_1|$, and the distance d is the hypotenuse.

$(AC)^2+(BC)^2=(AB)^2$

$|x_2-x_1|^2+|y_2-y_1|^2=d^2$

$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}=\sqrt{d^2}$

$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$



$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Ex: Find the *distance between (−2, 1) and (3, 4)*

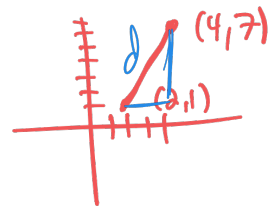
x_1, y_1 x_2, y_2

$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$d=\sqrt{(3--2)^2+(4-1)^2}$

$d=\sqrt{(5)^2+(3)^2}=\sqrt{25+9}=\sqrt{34}$

You try - (2, 1) and (4, 7).



$2^2+6^2=d^2$

$4+36=d^2$

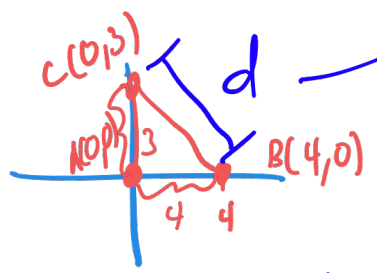
$d^2=40$

$d=\sqrt{40}=\sqrt{4\cdot10}=2\sqrt{10}\approx6.32$

Verifying a Right Triangle

Points A(0, 0), B(4, 0), and C(0, 3) are the vertices of a triangle. Find each side length, then use the **converse** of the Pythagorean Theorem to confirm it's a right triangle.

if $a^2+b^2=c^2$ Then a Right Δ



$3^2+4^2 \stackrel{?}{=} d^2$

$9+16 \stackrel{?}{=} d^2$

$25 = 25$ ✓

Right Triangle

(0,3) (4,0)
x_1,y_1 x_2,y_2

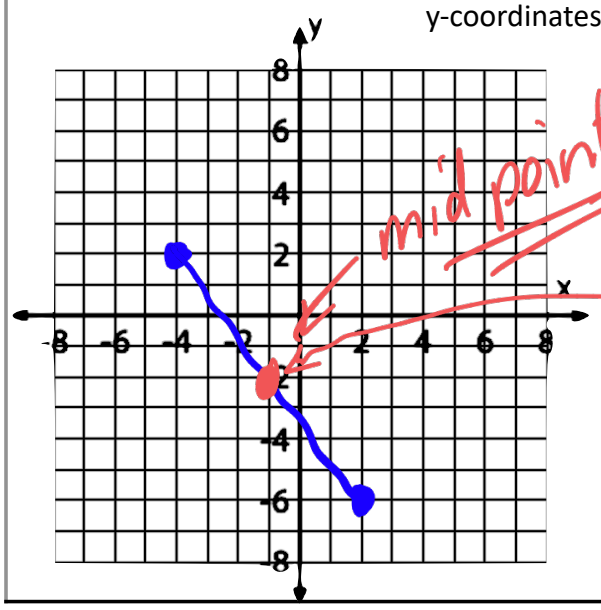
$d=\sqrt{(4-0)^2+(3-0)^2}$

$=\sqrt{16+9}$

$=\sqrt{25}=5$

The Midpoint Formula

The midpoint is just the average of the x-coordinates and the average of the y-coordinates.



$$M=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

Ex: Find the midpoint of the segment joining (−4, 2) and (2, −6).

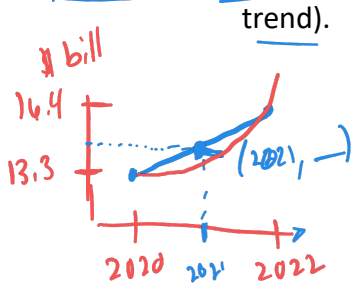
$M=\left(\frac{-4+2}{2},\frac{2+-6}{2}\right)=(-1,-2)$

You try —Find the midpoint of the segment joining (3, −1) and (−9, 5)

$M\left(\frac{3+-9}{2},\frac{-1+5}{2}\right)=(-3,2)$

Application: Estimating a Value

A company had sales of $13.3 billion in 2020 and $16.4 billion in 2022. Use the Midpoint Formula to estimate the 2021 sales (assume a linear trend).



(2020, 13.3)

(2022, 16.4)

$(2021, \frac{16.4+13.3}{2})$ 29.7

(2021, 14.85)

$ Billion

Section 1.1 Summary — Can you...

☐ Plot points and name their quadrant in the Cartesian plane?

☐ Use the Distance Formula to find the length between two points?

☐ Use the Midpoint Formula to find the center of a segment?

☐ Apply the coordinate plane to a real-life estimation problem?

Formulas at a Glance — Section 1.1

$d=|x_2-x_1|$ ✓

$a^2+b^2=c^2$ ✓

$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ ✓

$M=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$ ✓